

# Jiwon (Jesse) HAN

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## EDUCATION & APPOINTMENTS

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| POSTDOC | <b>Stanford University:</b> Stanford Science Fellow, Sept 2025 ~ Current                             |
| PHD     | <b>Harvard University:</b> Astronomy & Astrophysics, Sept 2025                                       |
| MSC     | <b>Harvard University:</b> Astronomy & Astrophysics, May 2022                                        |
| BSC     | <b>University of Virginia:</b> Physics-Astronomy, May 2019 w. Highest Distinction, Jefferson Scholar |

## RESEARCH INTERESTS

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Dark matter halo structure, near-field cosmology, hypervelocity stars, wide-field survey science.

## SELECT PUBLICATIONS

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- **Han, J. J.**, El-Badry, K., Lucchini, S., and 5 colleagues, *Hypervelocity Stars Trace a Supermassive Black Hole in the Large Magellanic Cloud*, The Astronomical Journal, 2025, 10.3847/1538-4357/adb967
- Lucchini, S. and **Han, J. J.**, *Threading the Magellanic Needle: Hypervelocity Stars Trace the Past Location of the LMC*, The Astrophysical Journal, 2025, 10.3847/2041-8213/ae109d
- **Han, J. J.**, Conroy, C., Zaritsky, D., and 4 colleagues, *Our Halo of Ice and Fire: Strong Kinematic Asymmetries in the Galactic Halo*, In Review for The Astronomical Journal, 2024, 10.48550/arXiv.2406.12969
- **Han, J. J.**, Conroy, C., and Hernquist, L., *A tilted dark halo origin of the Galactic disk warp and flare*, Nature Astronomy, 2023, 10.1038/s41550-023-02076-9
- **Han, J. J.**, Semenov, V., Conroy, C., and Hernquist, L., *Tilted Dark Halos Are Common and Long-lived, and Can Warp Galactic Disks*, The Astrophysical Journal, 2023, 10.3847/2041-8213/ad0641
- Cooper, A. P., Koposov, S. E., Allende Prieto, C., and 62 colleagues incl. **Han, J. J.**, *Overview of the DESI Milky Way Survey*, The Astrophysical Journal, 2023, 10.3847/1538-4357/acb3c0
- DESI Collaboration, Adame, A. G., Aguilar, J., and 262 colleagues incl. **Han, J. J.**, *The Early Data Release of the Dark Energy Spectroscopic Instrument*, arXiv e-prints, 2023, 10.48550/arXiv.2306.06308
- **Han, J. J.**, Conroy, C., Johnson, B. D., and 7 colleagues, *The Stellar Halo of the Galaxy is Tilted and Doubly Broken*, The Astronomical Journal, 2022, 10.3847/1538-3881/ac97e9
- **Han, J. J.**, Naidu, R. P., Conroy, C., and 9 colleagues, *A Tilt in the Dark Matter Halo of the Galaxy*, The Astrophysical Journal, 2022, 10.3847/1538-4357/ac795f
- **Han, J. J.**, *A Study on the Coffee Spilling Phenomena in the Low Impulse Regime*, Achievements in the Life Sciences, 2016, doi.org/10.1016/j.als.2016.05.009
- Lucchini, S., **Han, J. J.**, Hernquist, L., and 1 colleagues, *On the Origin of High-velocity Clouds in the Galaxy*, The Astrophysical Journal, 2024, 10.3847/1538-4357/ad6dde
- Lucchini, S., **Han, J. J.**, Hernquist, L., and 2 colleagues, *Invisible Accretion: Ionized Envelopes of TNG50 HVCs Can Sustain Star Formation*, The Astrophysical Journal, 2025, 10.3847/1538-4357/adf3b3

## SELECT WHITE PAPERS

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- **Han, J. J.**, Chiti, A., Chen, K.-F., and 93 colleagues, *A Path to an All-Sky Survey with Roman*, arXiv e-prints, 2026, 10.48550/arXiv.2602.21280
- **Han, J. J.**, Dey, A., Price-Whelan, A. M., and 206 colleagues, *NANCY: Next-generation All-sky Near-infrared Community survey*, arXiv e-prints, 2023, 10.48550/arXiv.2306.11784

## DISTINCTIONS & AWARDS

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|      |                                                          |                                                                         |
|------|----------------------------------------------------------|-------------------------------------------------------------------------|
| 2026 | Robert J. Trumpler Award                                 | Astronomical Society of the Pacific for outstanding PhD thesis          |
| 2025 | Eric R. Keto Prize                                       | Harvard-Smithsonian CfA ITC for best thesis in theoretical astrophysics |
| 2023 | Multidisciplinary Student Research Collaborative         | Harvard Radcliffe Institute                                             |
| 2020 | Bok Distinction in Teaching                              | Harvard College                                                         |
| 2019 | Limber Prize                                             | UVa award for best graduating student in Astronomy                      |
| 2017 | Ig Nobel Prize                                           | Award for scientific achievements that make people laugh, then think    |
| 2015 | Silver Medal, INTERNATIONAL YOUNG PHYSICISTS' TOURNAMENT | South Korea Delegate                                                    |

## MEDIA COVERAGE

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|----------------------------------|---------------------------------------------------------------------------------|
| <b>NPR All Things Considered</b> | “That galaxy next door? It’s home to a monster black hole” (March 6, 2025)      |
| <b>Forbes</b>                    | “Supermassive Black Hole Lurks In Nearby Large Magellanic Cloud” (Mar 08, 2025) |
| <b>Washington Post</b>           | “How to never spill your coffee, according to science” (Aug 29, 2016)           |
| <b>NPR Short Wave</b>            | “How To Keep Your Coffee In Its Cup: The Claw Grip” (Aug 13, 2019)              |

## TEACHING

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|             |                                                           |                   |
|-------------|-----------------------------------------------------------|-------------------|
| FALL 2024   | Cosmology (college senior level)                          | Harvard Astronomy |
| FALL 2021   | Galaxies & Cosmology (graduate level)                     | Harvard Astronomy |
| FALL 2020   | Astronomy 1 (college freshman level))                     | Harvard Astronomy |
| FALL 2018   | Astrophysical Processes (college senior & graduate level) | UVa Astronomy     |
| SPRING 2018 | Quantum Mechanics 2 (college senior level)                | UVa Physics       |
| FALL 2017   | Quantum Mechanics 1 (college junior level)                | UVa Physics       |

## SUCCESSFUL PROPOSALS

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| PI, <i>Chandra</i> Cycle 27:  | X-ray signatures of the LMC SMBH candidate (100 ks, 2025)                                          |
| Co-I, <i>Hubble</i> Cycle 33: | Proper motions of hypervelocity stars (7 orbits, 2025)                                             |
| PI, Magellan/LLAMAS:          | Introducing $(MC)^2$ : The Magellanic Clouds Magellan Campaign<br>(8 nights from Stanford and MIT) |
| Co-I, ATCA DDT:               | Radio counterpart of LMC SMBH candidate (14 hours, 2024)                                           |
| PI, VLT DDT:                  | Infrared characterization of LMC SMBH candidate (3 hours, 2024)                                    |
| PI, MMT:                      | Mapping Na D in High-velocity Clouds (14 nights, 2024)                                             |
| PI, VLA DDT:                  | Imaging High-velocity Clouds (13 hours, 2023)                                                      |

DDT = Director’s Discretionary Time

## MISCELLANEOUS

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|                                    |                                                                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Journal Refereeing:                | The Astrophysical Journal, The Astrophysical Journal Letters, Nature Astronomy, Monthly Notices of the Royal Astronomical Society, Astronomy & Astrophysics |
| Programming Languages:             | Python, C, C++, IDL, ROOT, MATLAB                                                                                                                           |
| La Serena School for Data Science: | Machine learning intensive school funded by NSF (Summer 2019)                                                                                               |
| CMS Data Analysis School:          | Particle physics data analysis training for CMS collaborators (Winter 2017)                                                                                 |
| Trail Work:                        | Overseer of the Southern District of Shenandoah National Park (2017 ~ 2019)                                                                                 |
| Climbing:                          | Ascents of 5.14 routes, V11 boulders, and WI5 ice routes.                                                                                                   |